

# From Interpretable to Black-Box: Dynamic Model Identification for Small USV Control Design

Brian S. Bingham and Carson Vogt  
Department of Mechanical and Aerospace Engineering  
Naval Postgraduate School, Monterey, CA, USA  
bbingham@nps.edu

Effective low-level feedback control of small underactuated uncrewed surface vehicles (USVs), specifically vessels under seven meters with a single thruster and rudder, requires dynamic models that provide both actionable design insight and predictive accuracy. Tow-tank testing is impractical at this scale; parameters must be identified solely from limited field trials. Existing system identification methods span a wide range of model complexity but offer little guidance on how much fidelity is enough, because sufficiency is only meaningful relative to a stated purpose. Absent that specification, the default is a bias toward more fidelity, since the benefit of an added term is readily quantified while its costs in identifiability, experimental burden and obscured design insight are not.

We propose a methodology built around a deliberate spectrum of three model levels, each identified from the same experimental data: (1) first-order, linear, single-input single-output (SISO) transfer functions, Nomoto-style models for surge and yaw that are immediately interpretable and support controller architecture intuition, but risk oversimplification when considered in isolation; (2) a physics-based nonlinear state-space formulation incorporating quadratic hydrodynamic drag; and (3) a Nonlinear AutoRegressive with eXogenous inputs (NARX) multilayer perceptron. Rather than seeking a single best model, the spectrum is used collectively: simple models anchor human intuition; structured nonlinear models expose which terms are dynamically salient; and the black-box model bounds the benefit of added complexity.

Preliminary results from step-response field experiments demonstrate the approach. First-order SISO models achieve  $R^2 = 0.93$  (surge,  $\tau = 1.95$  s) and  $R^2 = 0.83$  (yaw rate,  $\tau = 0.47$  s), providing direct controller tuning parameters. Cross-model diagnostics identify quadratic drag as the dominant surge nonlinearity and reveal that speed-dependent rudder effectiveness, a key coupling for underactuated control design, is not resolvable from a single short trial, motivating targeted data collection. A margin-based diagnostic makes the sufficiency question computable, reporting the forward speed at which the first-order model stops changing the control design. The NARX model matches one-step prediction accuracy ( $R^2 = 0.93$ ) but diverges in closed-loop simulation, demonstrating that physical structure provides essential regularization under limited data. A larger dataset spanning multiple operating regimes is being collected to evaluate the full methodology.